

Foundations of Machine Learning

IT582: 3-0-2-4

Instructor: Srimanta Mandal

Schedule

Lectures:

- Tuesday : 3:00 – 3:50 PM
- Thursday: 2:00 – 2:50 PM
- Friday: 4:00 – 4:50 PM

Lab:

Will be announced shortly.

Teaching

- The course Moodle is the place for every material.

 DA-IICT Courses

  Srimanta Mandal 

Foundations of Machine Learning

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Foundations of Machine Learning (IT582)

Your progress ?

Credit Structure: 3-0-2-4

Instructor: Srimanta Mandal (*srimanta_mandal@dau.ac.in*)

Grading Policy:

1st In-semester Exam	: 15%
2nd In-semester Exam	: 25%
End-semester Exam	: 35%
Lab Assignments	: 20%
Class Interaction	: 5%

 Announcements

Course Content and References

Brief Content:

Grading Policy (*Tentative*)

- 1st In-sem : 15%
- 2nd In-sem : 25%
- End-sem : 35%
- Lab Assignments : 20%
- Class Interaction : 5%

Lab Assignment

- Lab assignments will be given to you during lab sessions/in regular intervals.
- You need to implement the assignments during the lab session. (Use Python for implementation).
- Prepare a report about the assignment.
- You need to submit the report within a deadline.
- The assignments shall be evaluated in the lab schedule
- Lab assignments carry 20% marks.

Content in a Nutshell

- 1) *Regression* – Linear regression, Convexity and gradient descent algorithm, Vectorization, Feature Scaling, Concept of Regularization, Normal Equation, Probabilistic Linear Regression, Overfitting, Ridge regression, The Lasso, Basis Functions
- 2) *Resampling, Model Selection* –Generalization Error, Model Capacity, Cross-validation, The Bootstrap, Bias-Variance Decomposition
- 3) *Classification* – Logistic regression, Softmax, Linear Discriminant Analysis, Quadratic Discriminant Analysis, Decision trees, Boosting, Random forest, K-NN, Support Vector Machine
- 4) *Clustering* – K-Means, Gaussian Mixture Model
- 5) *Optimal Projection of Data*– Principal component analysis, Linear Discriminant Analysis
- 6) *Neural Network*- Representation, Forward propagation, Back Propagation

About the instructor



- Dr. Srimanta Mandal
- B.Tech: Electronics & Communication Engineering (2010) -- WBUT
- Ph.D.: Image Processing (2017) – IIT Mandi
- Post-doc (2018) – IIT Madras
- Area of Research: Image Processing, Computer Vision, Machine Learning
- More information on: <https://srimanta-mandal.github.io/>